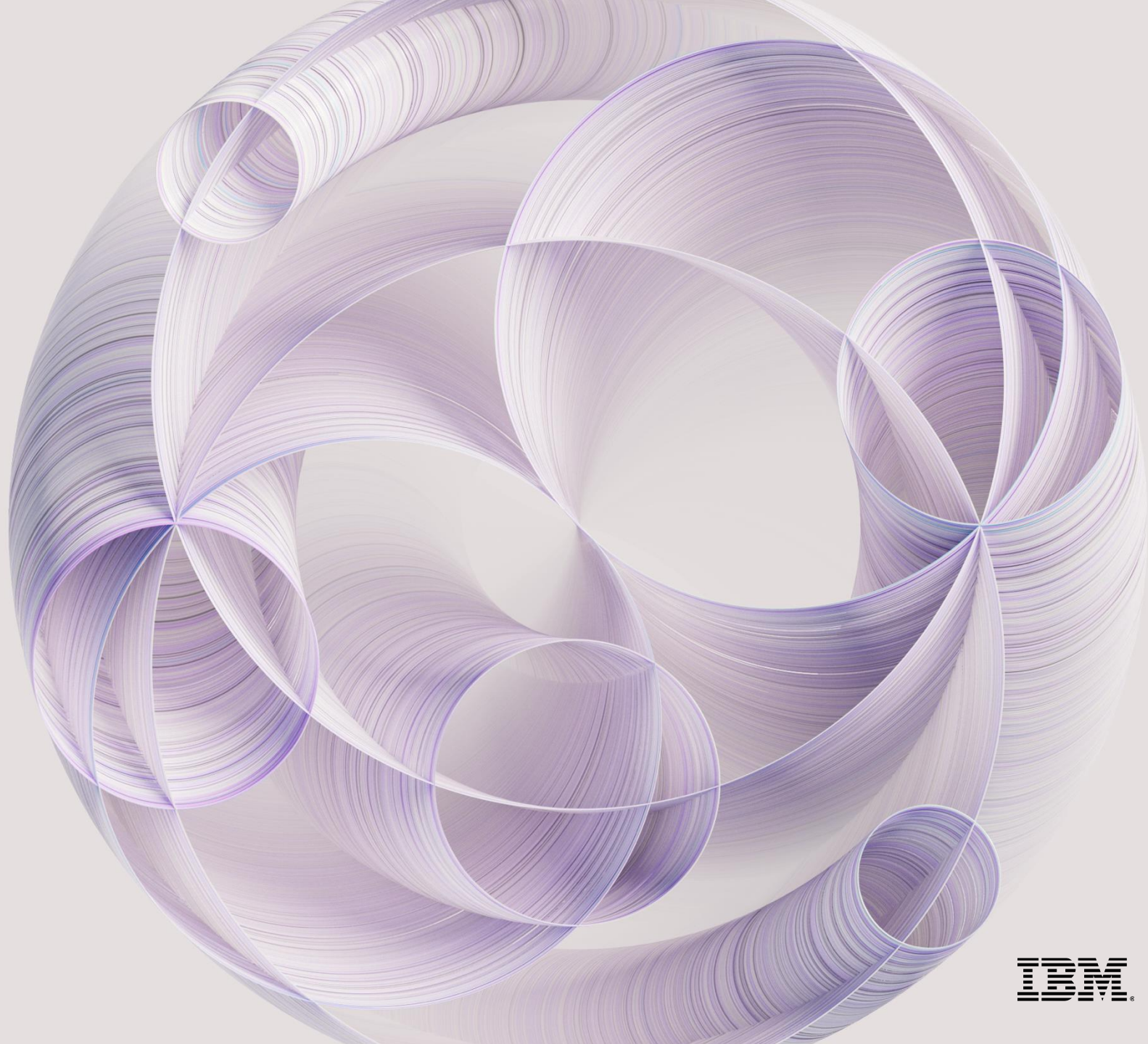


The Agent Journey: A Hands-On Path from Building and Integrating to Observability

AMLD 2026

Mohamed Ali Dhraief – AI Engineer
Dr. Naim Zierau – AI Architect
Philipp Gordetzki – AI Engineer



Your IBM Team



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AI Engineer

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Dr. Naim Zierau
AI Architect

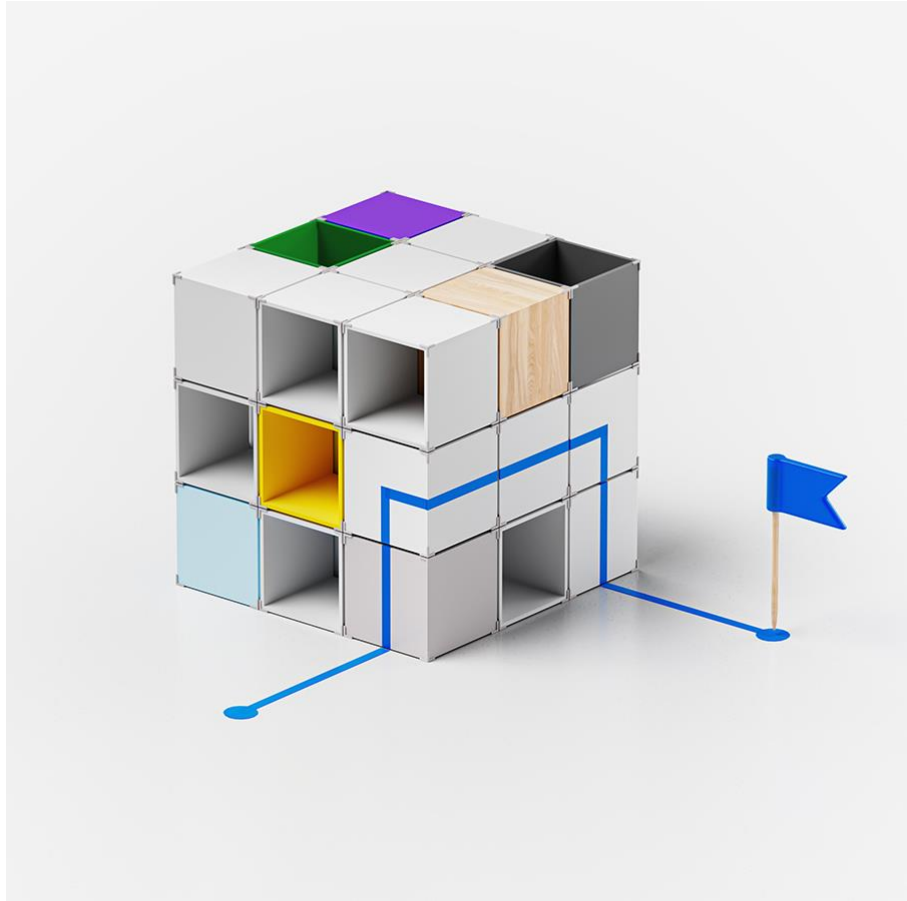
naim.zierau@ibm.com



Mohamed Ali Dhraief
AI Engineer

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Agenda



01

Sign up

02

Things to know before you start

03

Real-world example

04

Get your environment

05

Industry use case presentation
& workshop

06

5 common mistakes & pitfalls

07

Feedback & next steps

Setting the Scene

Setting up the Environment



https://ibm.biz/amld_ibm_workshop_sign_up

Setting up Expectations

This workshop is **not** about:

- Latest Models
- Research Trends

This workshop is about:

- Solving **Business** Problems with AI
- **How to move Beyond PoCs?**

Intro To Agentic AI



Things to know before
you start

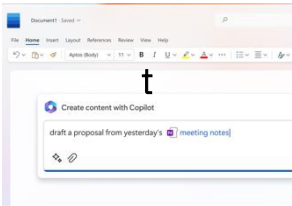
Generative AI today



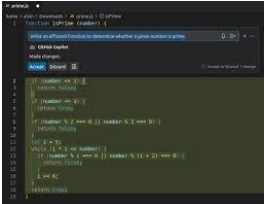
GenAI Status Quo

- Isolated, specific tasks
- Stand-alone solutions
- Hard to scale
-

RAG / Copilot



Code Generation



Challenges of “GenAI at Scale”

Individual, disjointed use-cases | limited process integration | high adaptation effort



Multiple agents working together

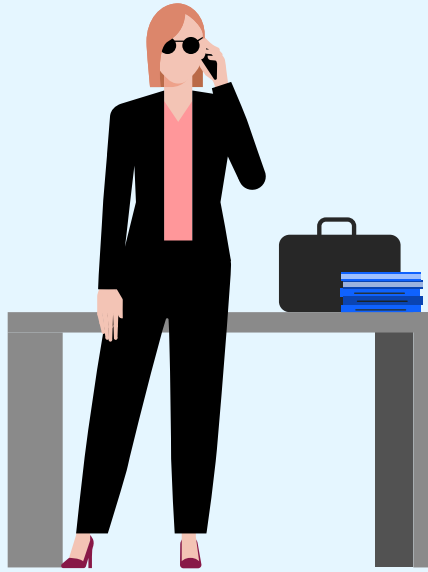
Automated, (BI-) integrated workflows

Modular Systems Engineering

Potential Solution: **Agentic AI?**

Agentic AI bot

Like a proactive personal agent



A **Gen AI-powered system** that not only understands and recommends but also **takes action** and completes tasks on your behalf, using **AI Agents**.



Hi Mike. I see that you clicked on our Premium Account promotion through social media. Are you interested in this account due to all the additional perks we offer? Or do you want me to show you all our similar options?

I'm looking for an account with unlimited transactions and additional perks, with the minimum fees.



Understood, then I recommend the Premium Checking Account. Would you like me to guide you through the application process right now?

Yes, please



Excellent! First, please provide your personal information. Next, we'll need to verify your identity. After that, we'll gather your employment and income information.

Once completed, your application will be reviewed, and you'll receive a confirmation within 3-5 business days. Let's get started!



How does each Agent know what to do?



Behaviour

It's specified **how the agent responds to user requests** and workflows. It defines the logic and behaviour, handling tasks, decisions, and interactions.



Tools

The agent can be provided with **functional capabilities** like API access, data processing, AI services or other Agents.



Knowledge

Files or content repositories enhance the Agent capabilities to generate **accurate and contextual responses**.

f.e. pdfs, word, etc... that the Agent can use to answer user's questions.

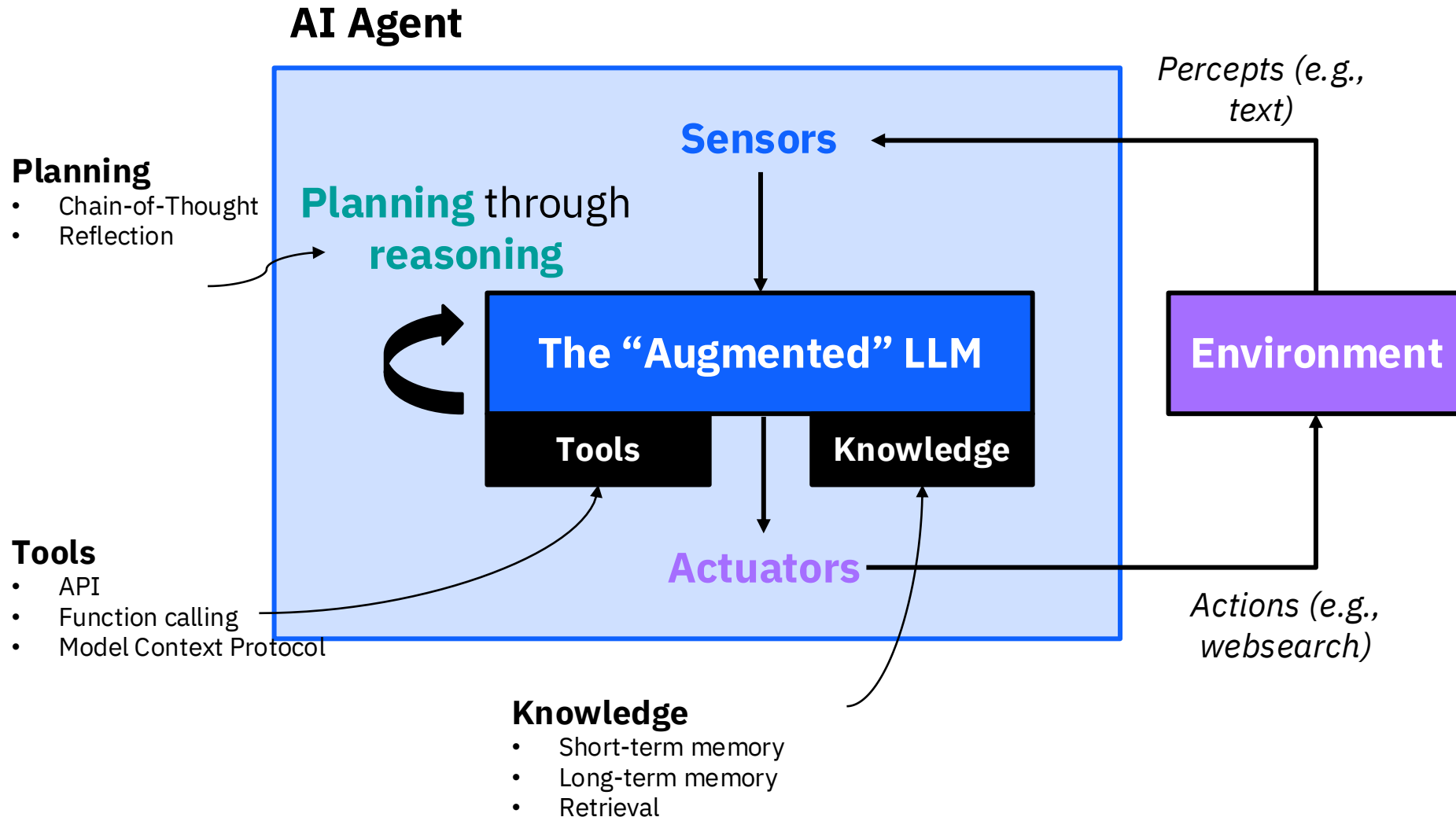


Channels

The agent can be **connected to interfaces** through which the **agent communicates**.

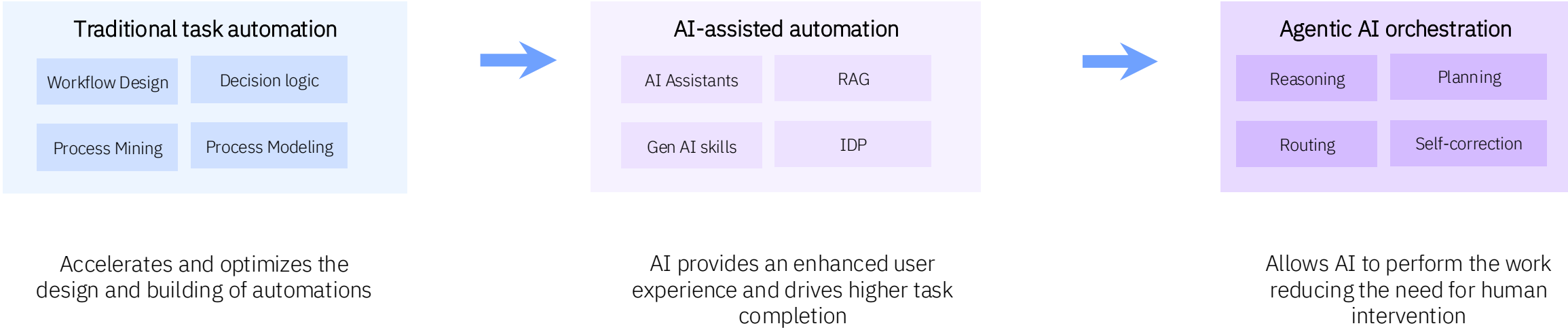
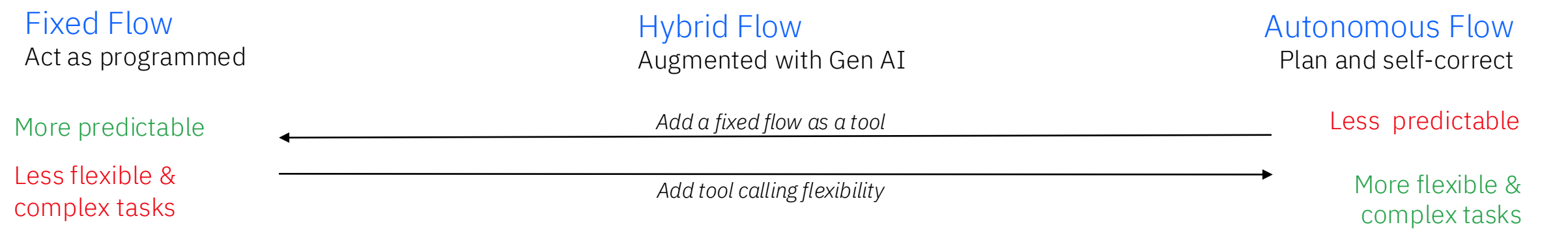
It integrates your agent with platforms to enable seamless user interaction through widely adopted communication channels.

AI Agent: Autonomously Performing Tasks by Designing its own workflow and utilizing memory and tools



An **AI agent** is a system that is capable of **autonomously performing tasks** on behalf of a user or another system by **designing its workflow** and utilizing available **tools**.

While the future is agentic AI, enterprises need flexibility in the approach they take across workflows

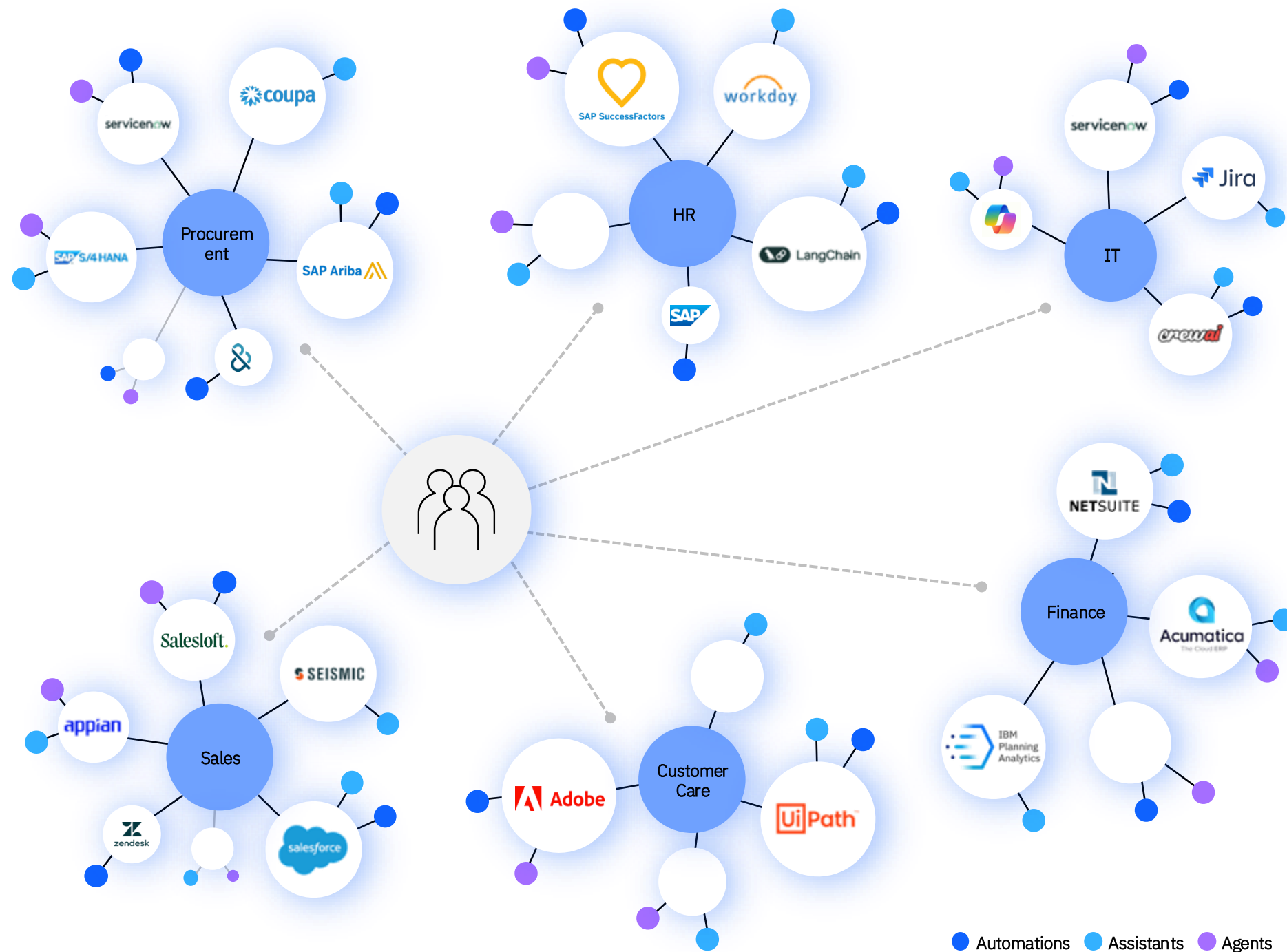


AI agents are going to be everywhere increasing complexity

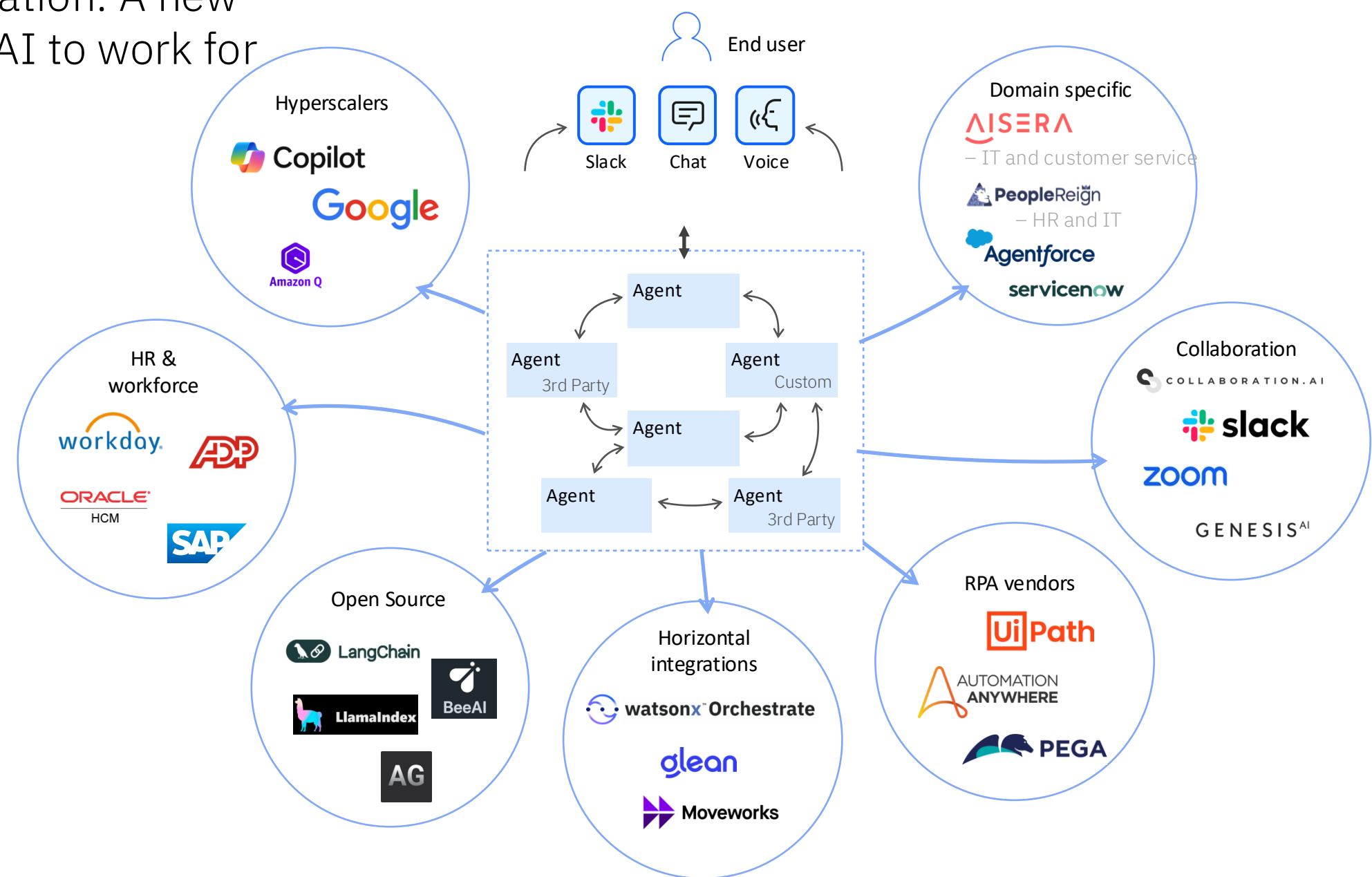
Breaking down silos

Taming agent sprawl

Bridging the production gap



Agentic Orchestration: A new paradigm to put AI to work for the enterprise



Orchestrate everything

How can I help? ▶

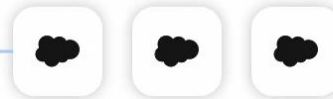
Agentic AI
Orchestration

Custom own agent

Build your own agent

3rd party agents

Your data



Your tools



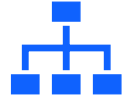
Your workflows



Your assistants

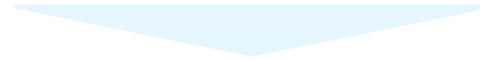


Overview of Agentic AI Use Cases in Switzerland



Enterprise Agents

Unified access of agents and tools across platforms, frameworks, and applications



Domain Agents



Automation Agents

Get Back-Office Tasks Done

- Customer Communication Processing
- Claims Processing
- Contract Lifecycle Management
- ...



Employee Agents

Support employees to be more effective

- Budget Assistant
- KYC Guide
- AskHR, AskIT
- ...



Customer Agents

Free customer agents to focus on value

- Personal Banking Assistant
- Omnichannel Appointment Agent
- Voice-based Card Blocking Agent
- ...



Enterprise Capabilities

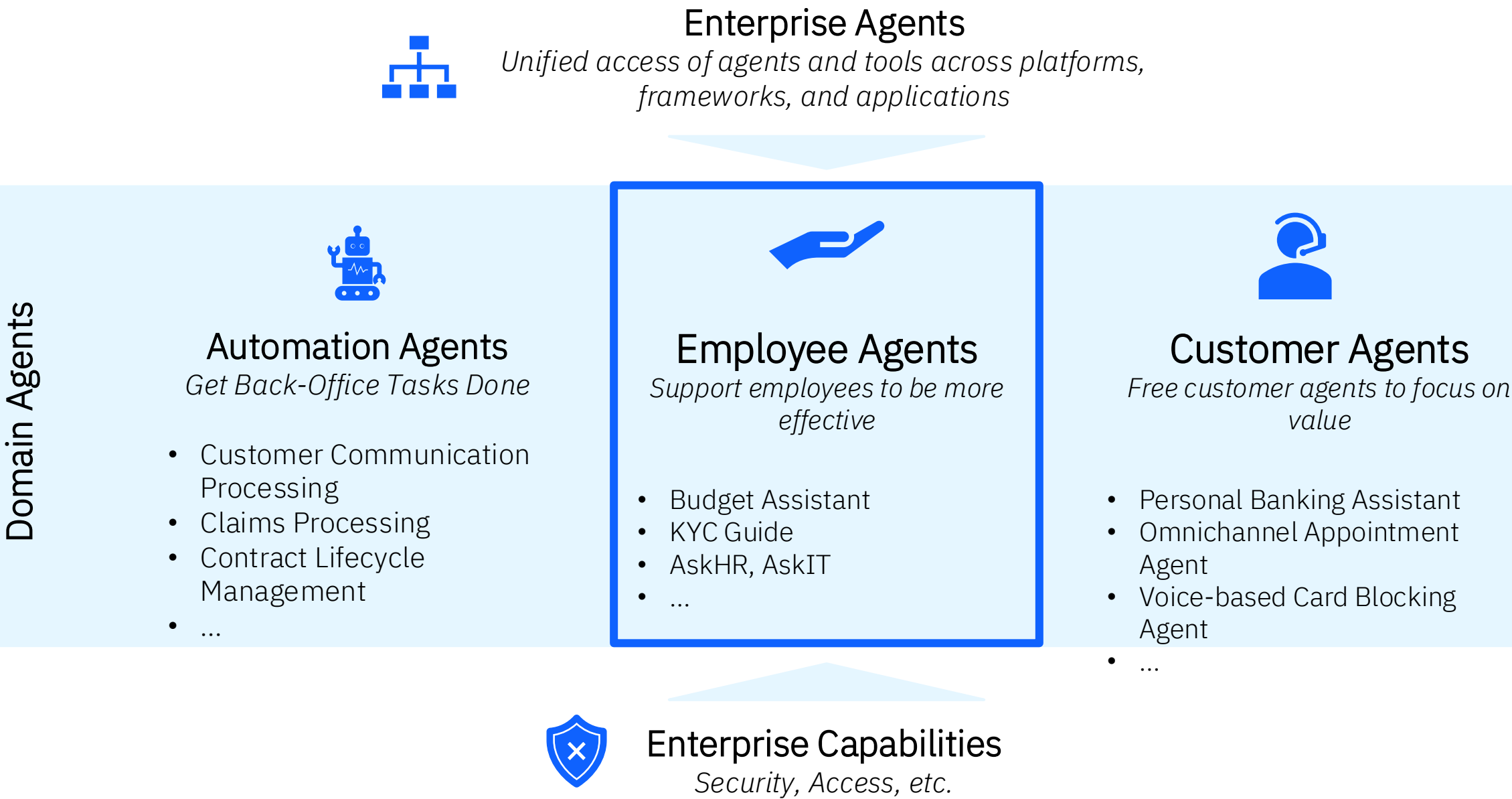
Security, Access, etc.

Industry Use Case Presentation



Budget Assistant

Overview of Agentic AI Use Cases in Switzerland



Budget Assistant

Key business aims

On-Demand Financial Insights:

Executives and department heads can access key data anytime, anywhere within Microsoft Teams.

Accelerated Decision-Making:

Finance teams spend less time on manual reporting and more on analysis and strategy.

Seamless Integration:

watsonx Orchestrate connects Teams and Planning Analytics, eliminating tool switching and data silos.

Martin's agentic AI journey

01 Natural Query:

Martin asks for financial planning data directly in **Microsoft Teams**.

02 Smart Routing:

The request is processed by **IBM watsonx Orchestrate**.

03 Automated Execution:

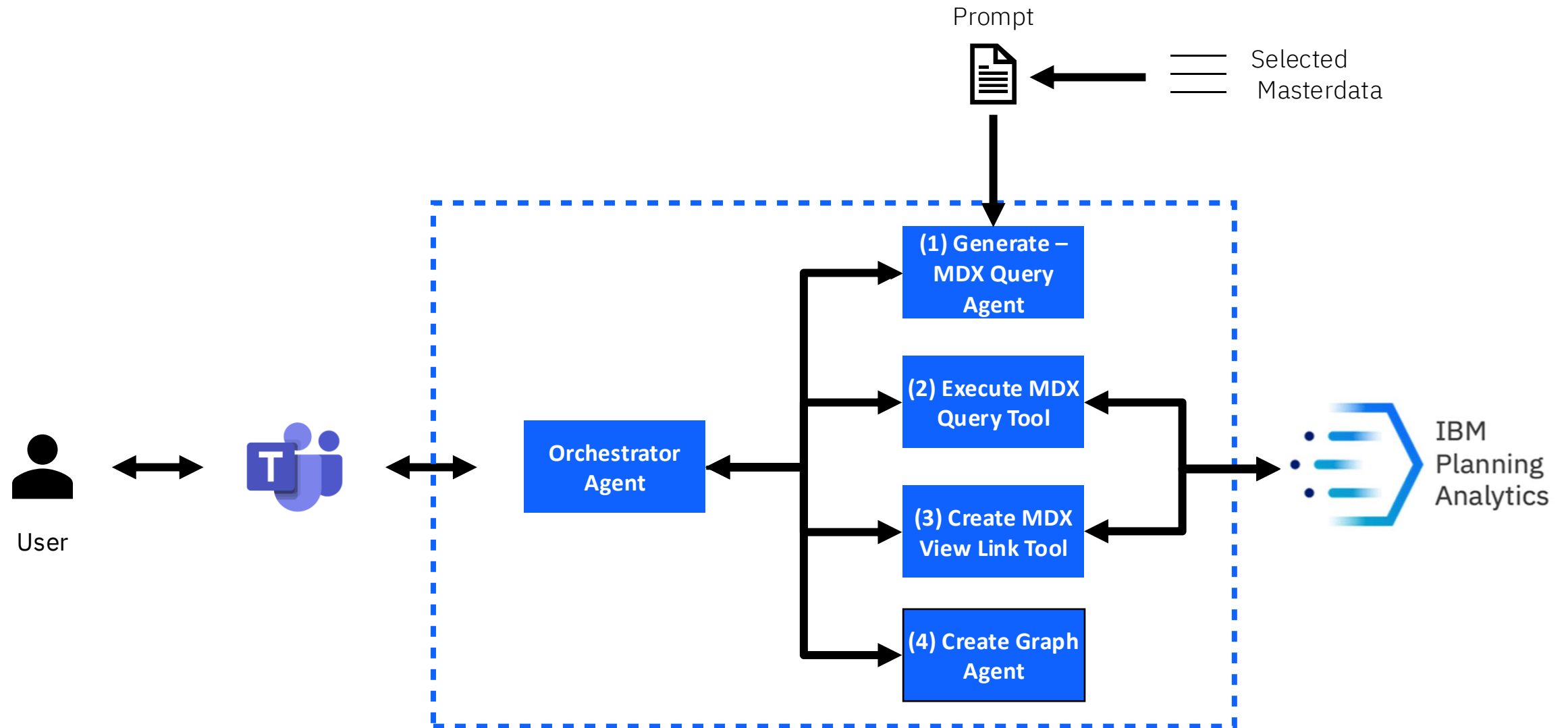
Orchestrate translates the query into **MDX**, runs it in **IBM Planning Analytics**, and generates a link to the relevant analytics cube.

04 Instant Results:

Insights are returned to Martin in Teams — no app switching, no delays.



Budget Assistant: Solution Overview

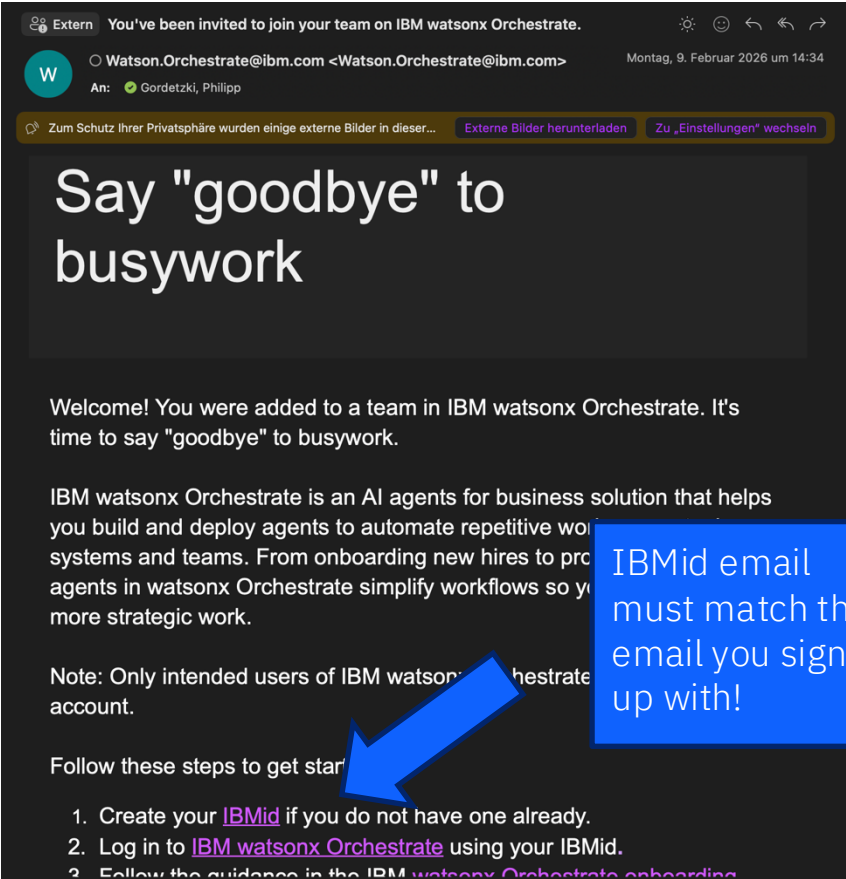


Hands-on lab

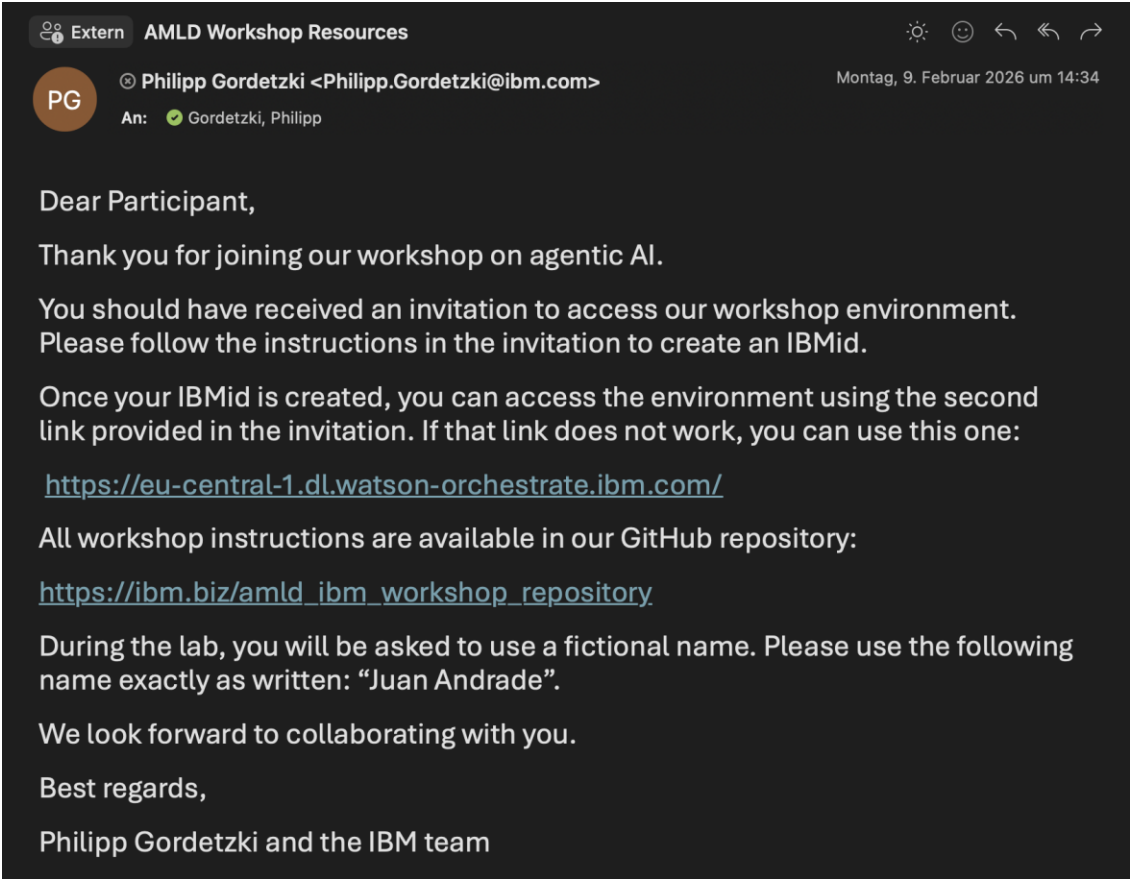


You received two emails from us

Invitation to Orchestrate



Email with links and fictional "name"



Workshop set up

1

Create your IMBid (ensure emails match)

2

Log in to Orchestrate

- Link in first email might be blocked → use link in second email
- If prompted to 2FA, “send code to email” is easiest
- Keep tab open

3

In a new tab / window: Open GitHub repository

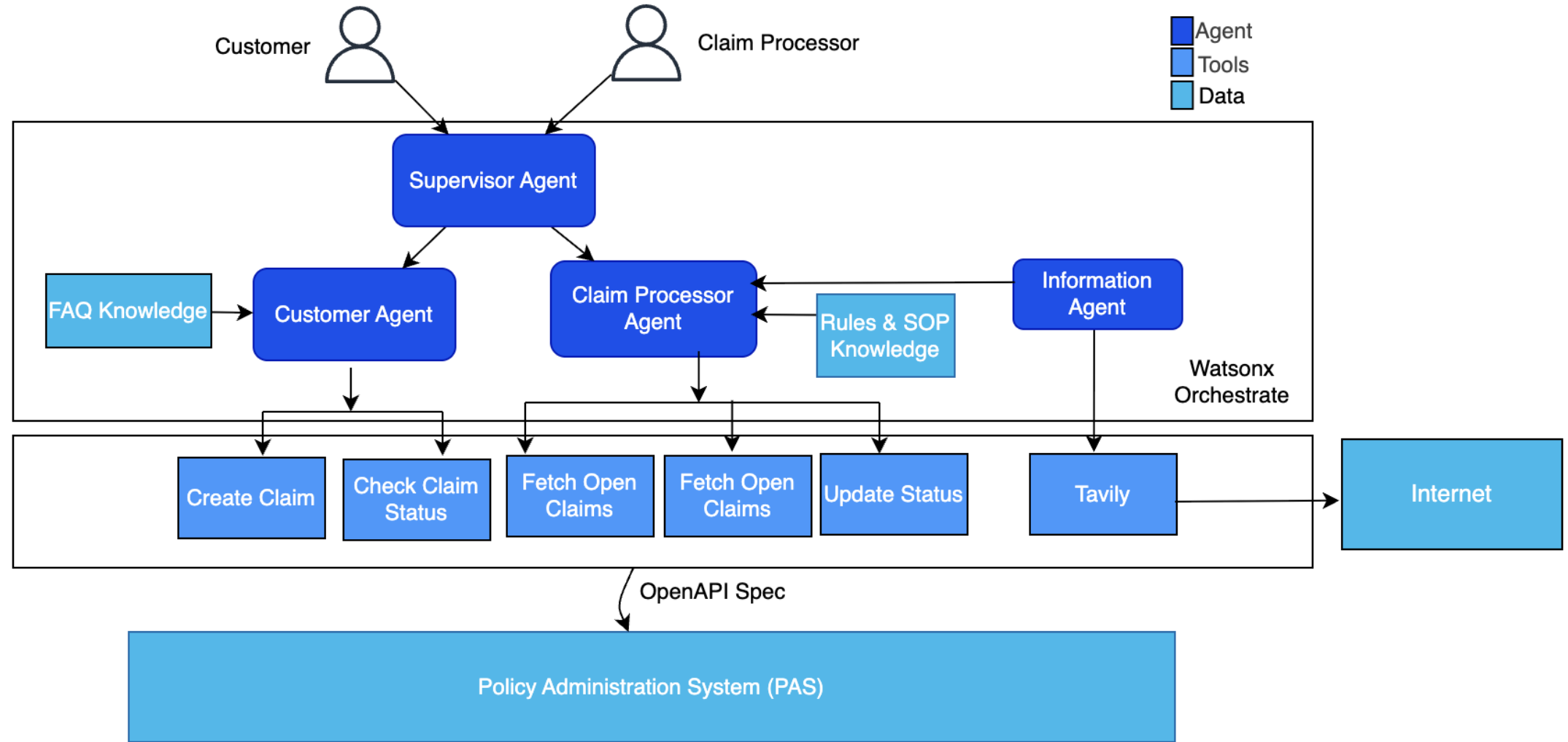
- Link in second email

GitHub repository



https://ibm.biz/amld_ibm_workshop_repository

Architecture overview



Common Mistakes & Pitfalls



How to address them?

Common Industry Challenges & Pitfalls



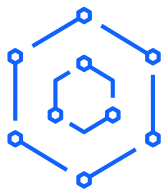
Integration

Balance cost and performance by having flexibility in the design of your Agentic AI architecture across the Enterprise



Security & Access Management

Manage agent operations with confidence through audit trails, analytics, HAP filtering etc



Governance

Governed Catalog of Agents and Tools are evaluated and deployed so they can be reusable across the organization



Explainability and transparency

Transparent evaluation of AI and AI Agents to ensure accuracy, answer correctness & instruction adherence



Accelerated time to value (Business Case)

Single studio with pre-built Gen AI and Automation capabilities to accelerate agent development for business and technical builders



Everywhere

Deploys natively across multiple clouds or on-prem

Common Industry Challenges & Pitfalls



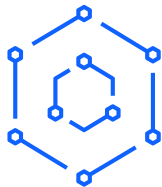
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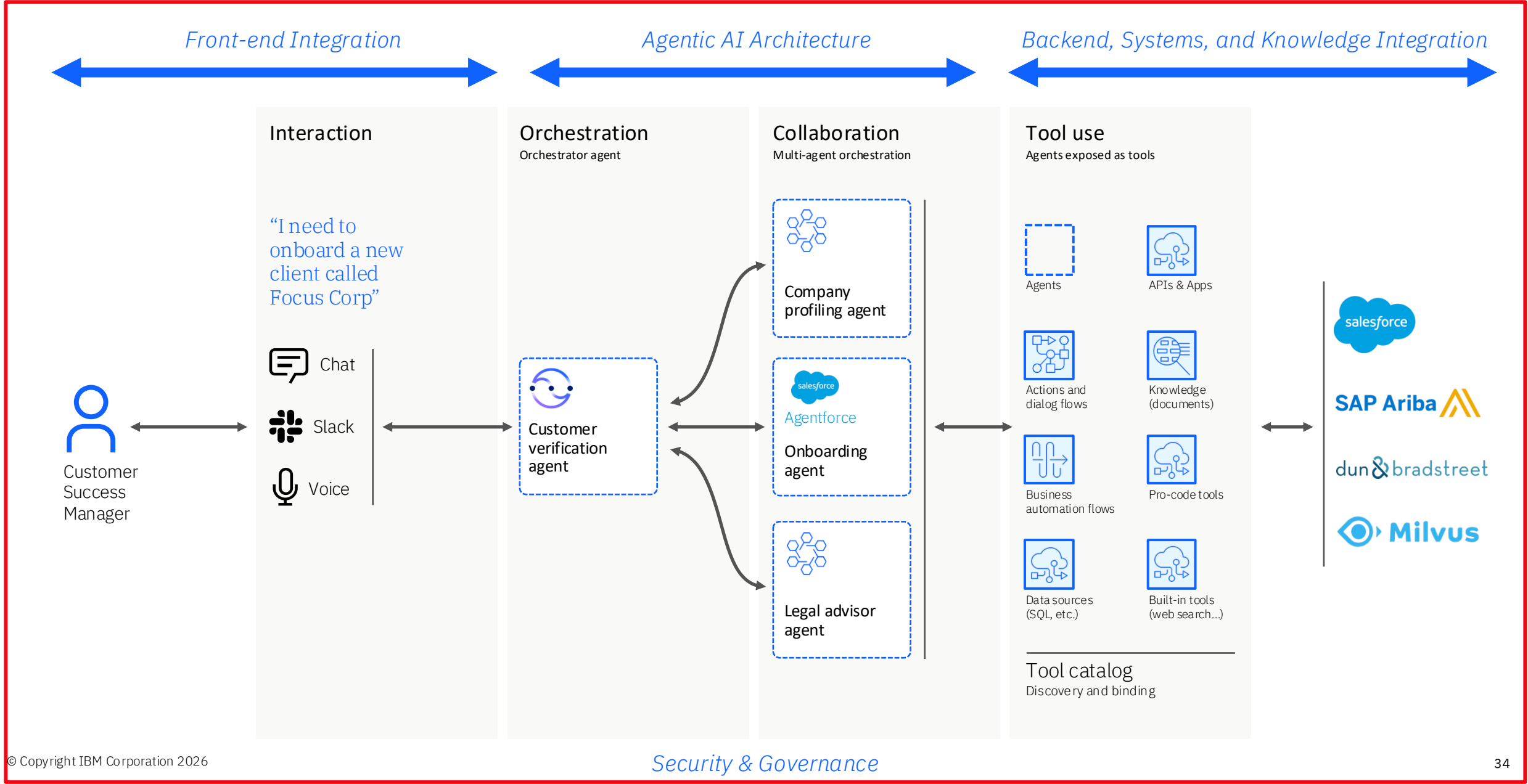
Single studio with pre-built Gen AI and Automation capabilities to accelerate agent development for business and technical builders



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Platform approach to multi-agent architecture



Common Industry Challenges & Pitfalls



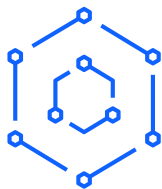
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Over-Agentication

There is such thing as too many agents.



Business Case

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Everywhere

Deploys natively across multiple clouds or on-prem

Enterprise-grade security

Enterprise security starts with role-based access control pass-through, ensuring every user is assigned a governed role that strictly limits access to only the data, tools, and actions they're authorized to use



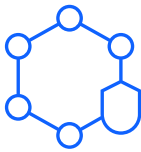
Manage security & governance

AgentOps & governance solutions



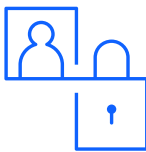
Authentication

- Support for different authentication schemes
 - Basic authentication
 - API key-based authentication
 - Bearer token authentication
 - Key-value pair authentication
 - OAuth 2.0 variants



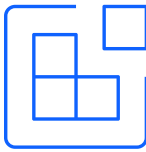
Credentials

- Credentials for teams and individuals
- Secure credentials
 - Encrypted
 - Isolated
 - Integrated with external credential management systems
- Secure networking
 - IBM Cloud Satellite connectors
 - Secure VPN configurations
 - Private IP access
 - IP whitelisting



Authorization

- Federated identity and single-sign-on
 - Continuity of user context across systems
 - Consistent authorization enforcement
 - Full traceability of agent actions as extensions of the authenticated user



Compliance and isolation

- Data isolation offering
- Compliance
 - SOC2, ISO27K, PCI, HiPAA

Common Industry Challenges & Pitfalls



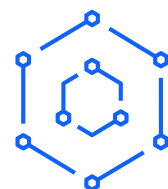
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Security

Manage agent operations with confidence through audit trails, analytics, HAP filtering etc



Everywhere

Deploys natively across multiple clouds or on-prem

01

Evolving elements of AI risk



New

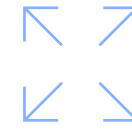
Emerging areas *intrinsic* to agentic AI

Risks

- Unsupervised autonomy
- Data bias
- Redundant actions
- Attack on AI agent's external resource
- **Tool choice hallucination**
- Sharing IP/PI/confidential information

Challenges

- Reproducibility
- Traceability
- Attack surface expansion
- Harmful and irreversible consequences



Amplified

Known areas *intensified* by agentic AI

Risks

- **Misaligned actions**
- Discriminatory actions
- Over- or under-reliance
- Unauthorized use
- Exploit trust mismatch
- Unexplainable or untraceable actions
- Lack of transparency

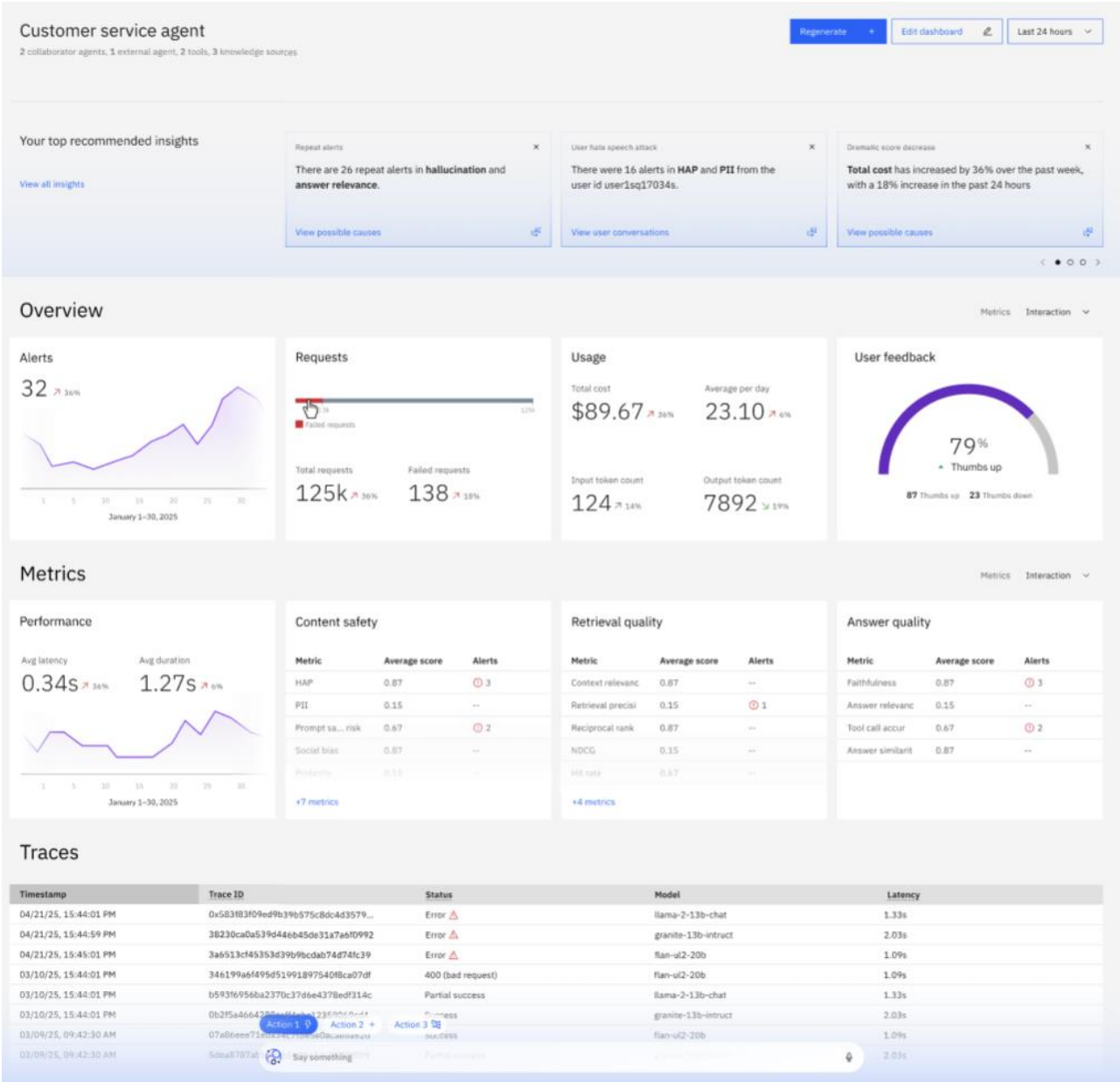
Challenges

- Evaluation
- Accountability
- Compliance
- Mitigation and maintenance
- **Infinite feedback loops**
- Shared model pitfalls

02

AgentOps

Observe, evaluate and optimize secure, compliant agents that are built anywhere, running everywhere.



Common Industry Challenges & Pitfalls



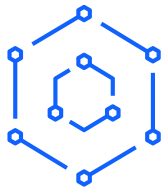
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Accelerated time to value (Business Case)

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Everywhere

Deploys natively across multiple clouds or on-prem

While the future is agentic AI, enterprises need flexibility in the approach they take across workflows



Balance flexibility and reliability

Combine GenAI & deterministic

Fixed Flow

Act as programmed

Hybrid Flow

Augmented with Gen AI

Autonomous Flow

Plan and self-correct

More predictable

Less predictable

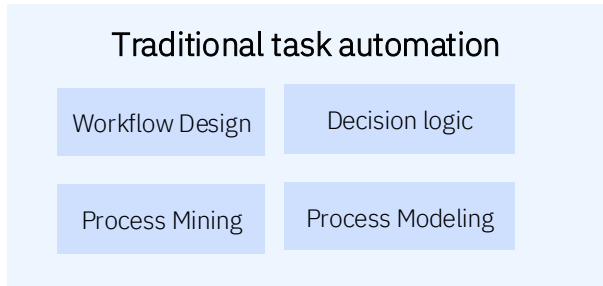
Less flexible & complex tasks

More flexible & complex tasks

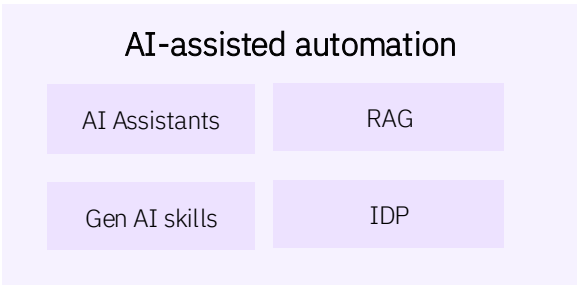
Add a fixed flow as a tool

Add tool calling flexibility

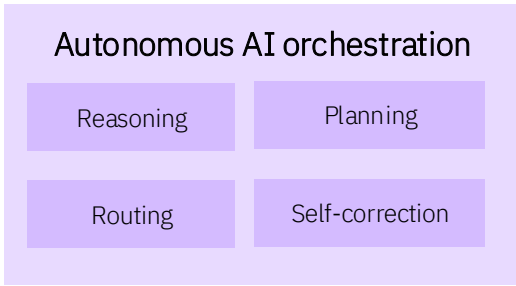
43



Accelerates and optimizes the design and building of automations



AI provides an enhanced user experience and drives higher task completion



Allows AI to perform the work reducing the need for human intervention

Common Industry Challenges & Pitfalls



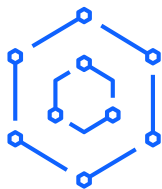
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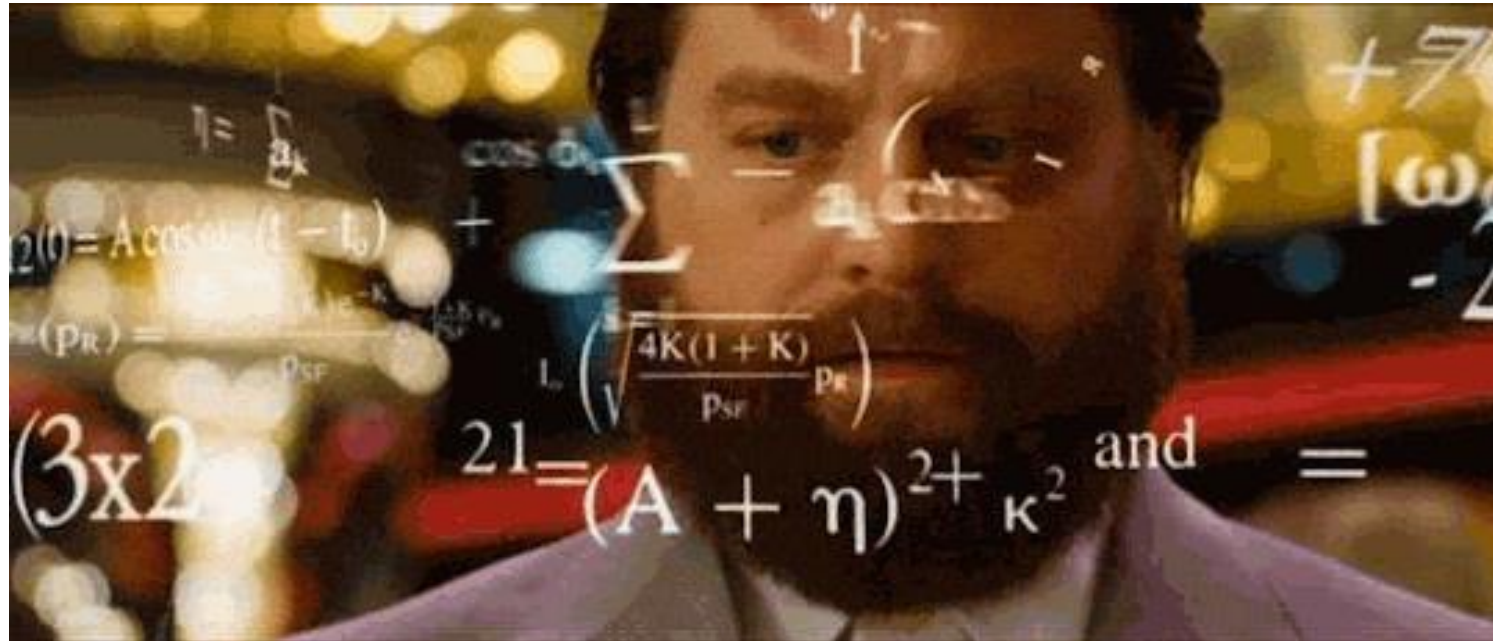
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Deploys natively across multiple clouds or on-prem

Getting Started

From Use Case to Business Case

From the thousands Agentic AI Use Cases out there...how do you identify the financially viable ones?



Value Creation

From Use Case to Business Case



Business
Opportunity
Statement

Or

Pilot Statement



Measure
Impact

- Goals
- Metrics
- Current Process
- Outcomes
- Stakeholders



Value
Tree

- Core Value Drivers
- Enablers and Constraints
- Interventions or Levers
- Value Outcomes



KPI Calculations
and Data Needs

- KPI Calculations and Data Needs
- Defined KPIs and formulas
- Data sources and Frequency

Common Industry Challenges & Pitfalls



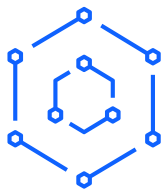
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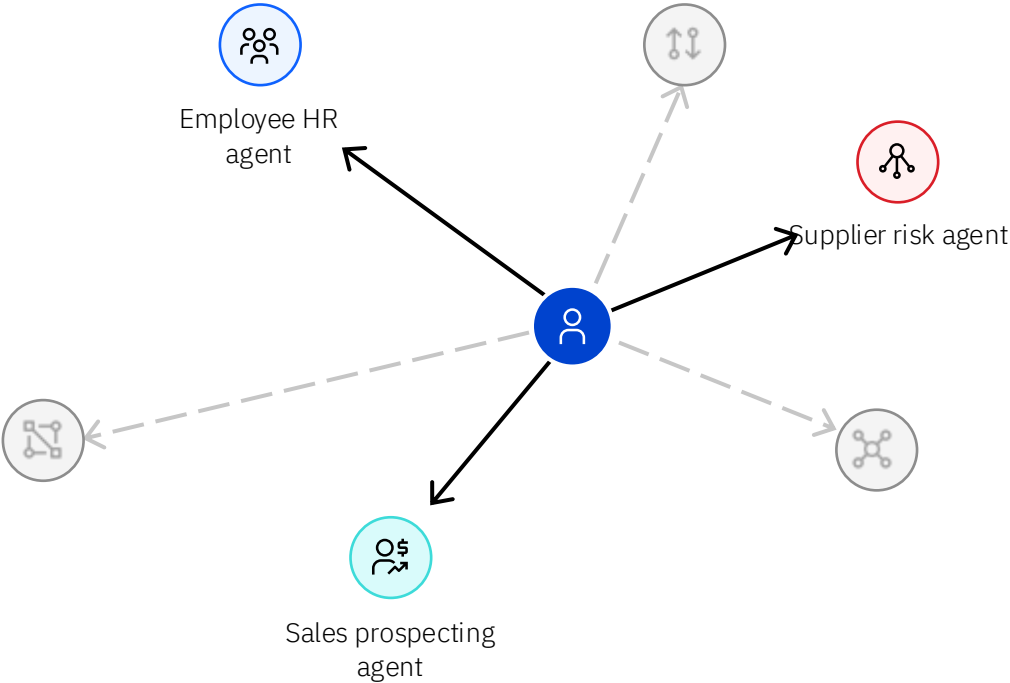


Orchestration

Offers a single point of entry for your users.

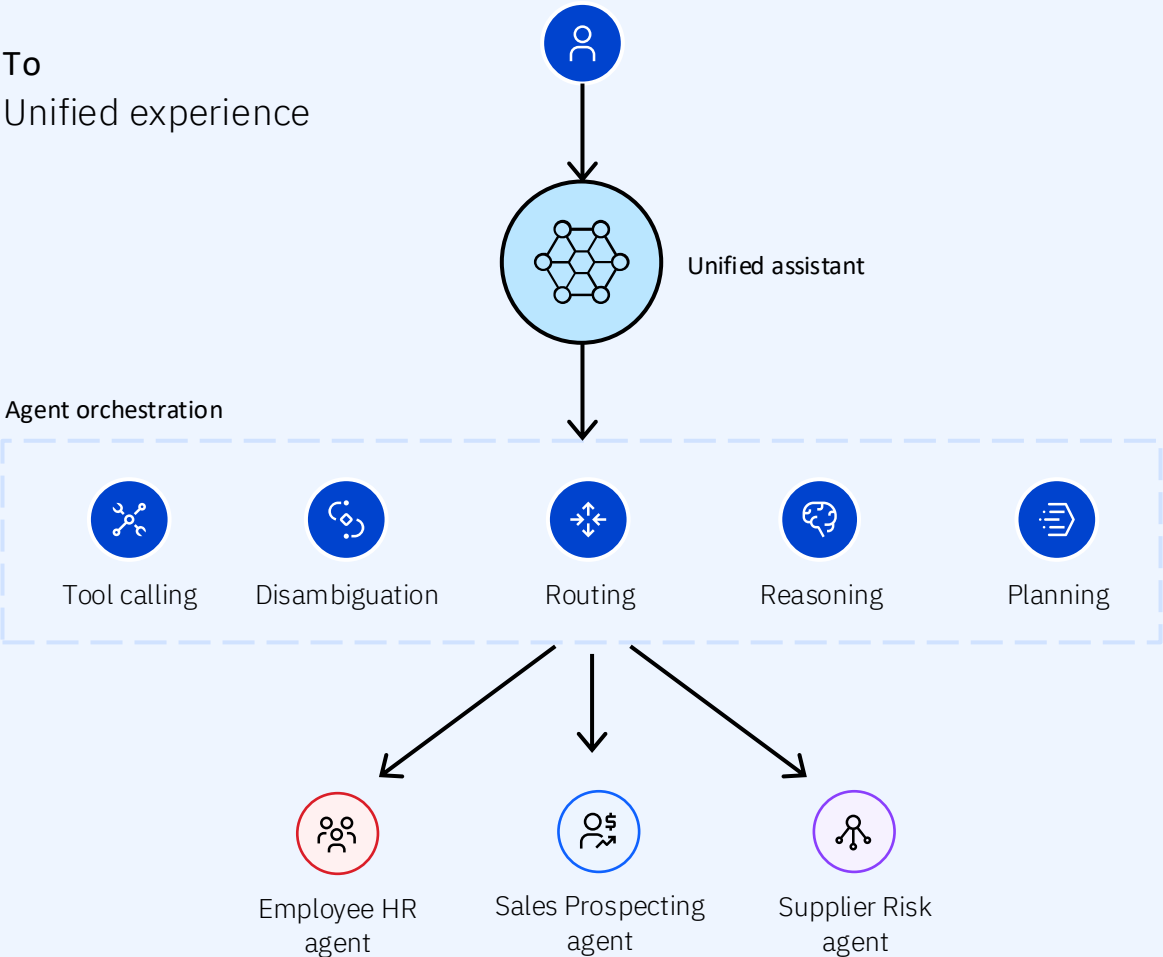
Multi-agent orchestration from fragmented to unified

From
Multiple entry points, fragmented experience



No orchestration

To
Unified experience



Agentic orchestration

Client zero: The path to \$4.5B (and counting)
AI productivity gains since 2023

AskIBM

A central unified interface for all IBMers, connecting to each domain assistant

IBM AskHR

10M

Annual HR interactions fully resolved by AI

40%

Reduction of HR operating budget

+55

Improvement of HR NPS score

IBM AskIT

80%

Inquiries resolved via AskIT

100

Days to build + deploy AskIT from scratch

50%

Reduction in support tickets after deployment

IBM AskSales

180K

Hours per week saved in gathering account insights

5,000

Seller questions answers per week

40%

Improvement in quality of outreach content

IBM Procurement

85%

Orders now processed via 'touchless procurement'

50%

Reduction in time spent on manual, repetitive tasks

15%

Enterprise workforce comprised of contractors

IBM AskIncentives

96%

Sales related inquiries contained within AI Assistant

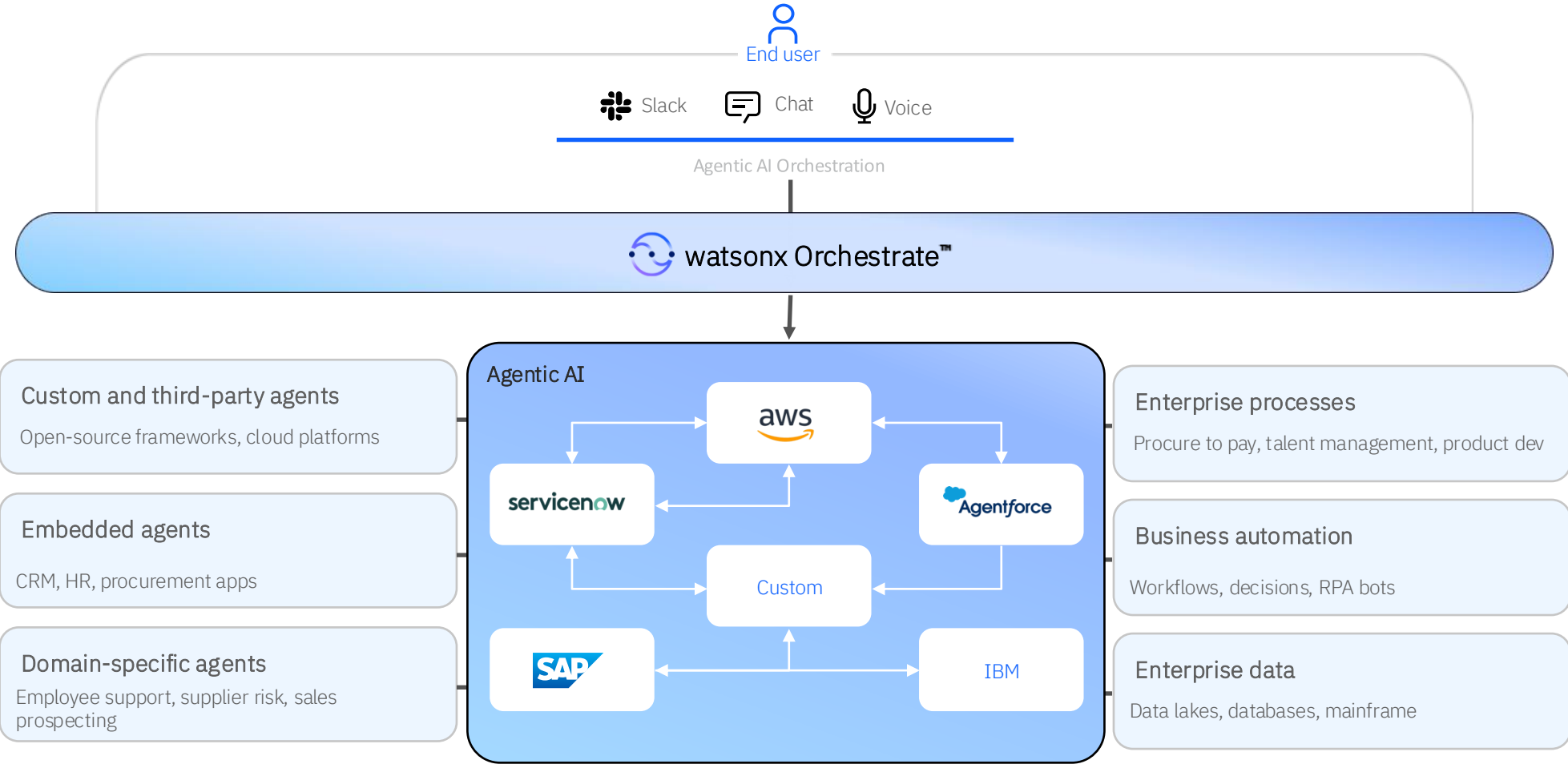
90%

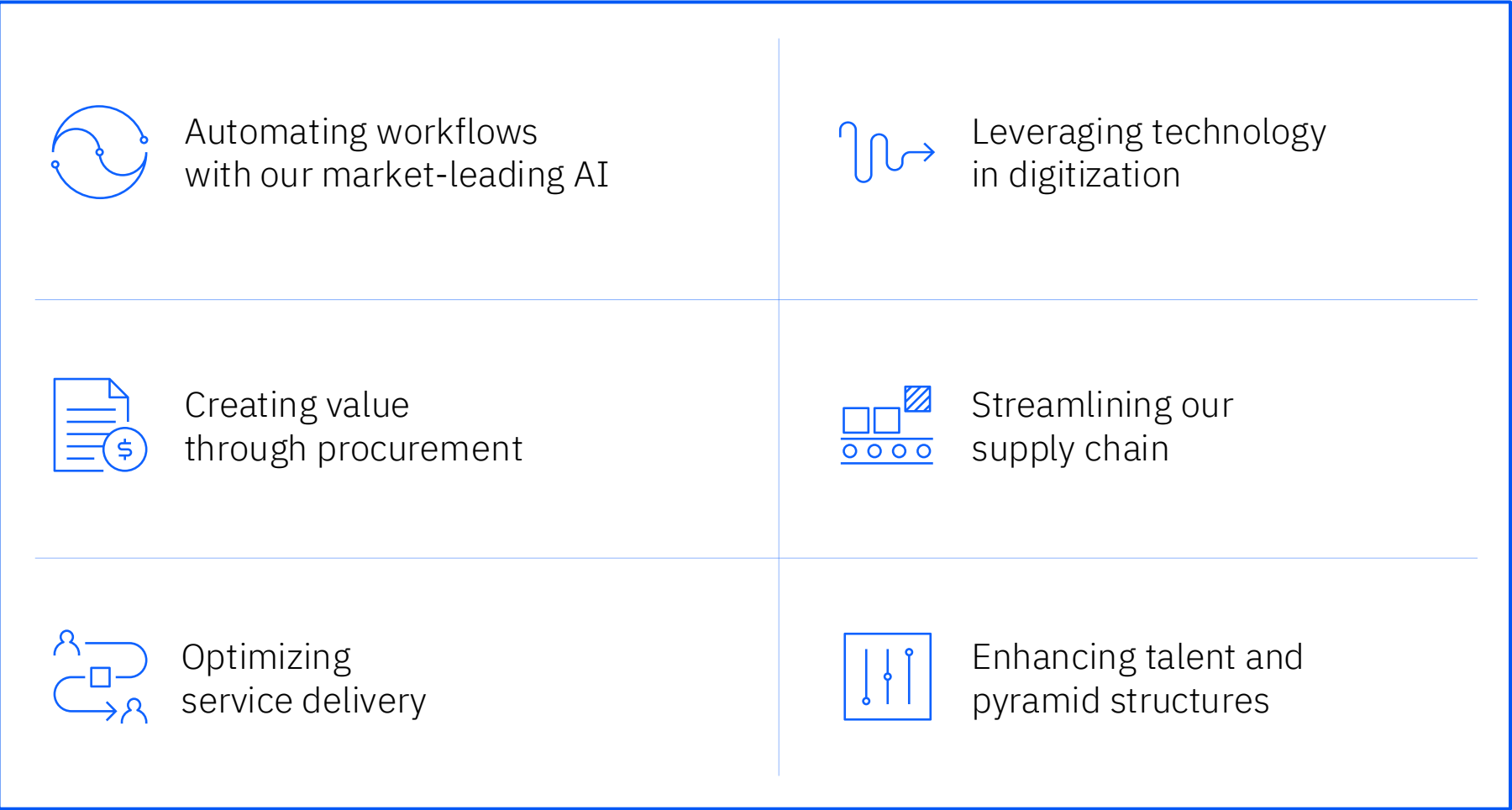
Greater accuracy in expense accruals

76%

Increase in productivity while serving 22k sellers

Multi-agent orchestration
across vendors and frameworks





Initial goal:

\$2B

Productivity savings 2024 exit run rate



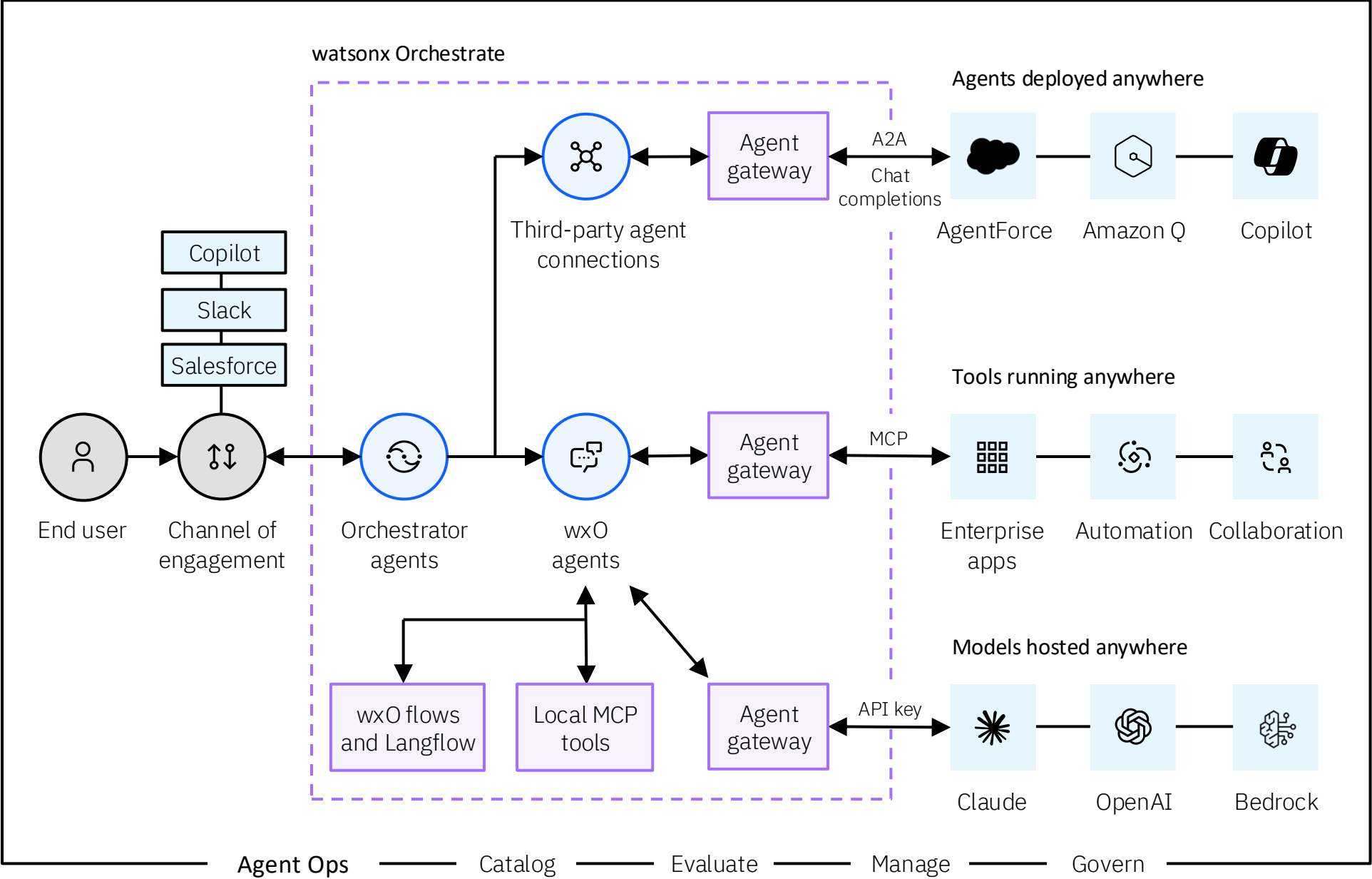
Impact:

\$4.5B

Productivity savings 2025 exit run rate

Open agent framework

Open, interoperable and pluggable agentic orchestration with centralized governance for agents running anywhere



Feedback & Next Steps

AMLD Feedback Form





watsonx Agentic AI Bootcamp

Build and deploy an Agentic AI application prototype for an enterprise use case

- Learn the top Agentic AI skills and best practices from our team of experts
- Leverage our watsonx tools for low-code fast prototyping of Agentic workflows
- Trust your agents with guardrails

Use cases



HR Automation

Build an HR Assistant using your enterprise data and documents to react and respond to employee inquiries



Intelligent Assistant

Train an assistant for customer service support and issue resolution



Business Automation

Utilize agents to automate a core business process through structure, discover, deconstruct, and develop.



Bring your Own Usecase (2nd day)

Bring to life your own use case that generates business value to your organization with the help of our team of AI experts

Accenture report says “enterprises are beginning to embrace AI agents, noting that 77% of executives agree AI agents will reinvent how their organization builds digital systems. Additionally, 48% of executives say AI agents would improve flexibility of their organization’s digital architecture”

[Read the Accenture Report](#)

Agenda

1. **Learn:** develop your skills and best practices on AI Agents (tailored to skill level)
2. **Use case & data definition:** bring to life your own use case or select one from our list of top industry use cases
3. **Implementation:** apply your skills hands-on to build an Agentic AI prototype alongside with our AI experts
4. **Deployment:** demonstration of how it would look in real life and the trust considerations needed to go to production

Explore the value

Audience: AI Engineers / Line of business participants

Format: in person workshop

- **1-day:** learn and implement a predefined use case
- **2-days:** learn and bring to life your own use case

Outcomes:

- A functional prototype of an Agentic AI application to address your enterprise use case
- The opportunity to continue to work with our AI experts in a longer-term co-creation pilot